KATHMANDU UNIVERSITY

SCHOOL OF ENGINEERING

DEPARTMENT OF MECHANICAL ENGINEERING

LABORATORY REPORT ON

MEEG 404 LABORATORY WORKS

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VIBRATION MEASUREMENT OF THE TURBINE

AIM

- To play with a vibration measurement instrument.
- To detect faults in turbine components using vibration level.

MATERIAL REQUIRED

1. Piezoelectric Transducer
2. Data Acquisition System
3. Vibration Module
4. Laptop with Lab View and Python software

THEORY

Turbines shake and vibrate because of different forces, like uneven weight, parts not lining up properly, worn-out bearings, or the way air or water moves through them. Measuring these vibrations is important to keep the turbine running smoothly and safely.

To check vibrations, special sensors are used. These sensors record how much the turbine shakes and help find possible problems. The data collected is often converted into a frequency-based format using a method called Fast Fourier Transform (FFT), which makes it easier to spot issues.

There are different ways to measure vibration:

- **Acceleration** tells how fast the vibration changes.
- **Velocity** shows how strong the vibration is.
- **Displacement** measures how far parts move when vibrating.

By studying these vibrations, experts can detect problems early, like worn-out parts or misalignment. There are rules, like **ISO 10816**, that set safe vibration levels. If vibrations go beyond these limits, it means the turbine might need repairs.

Keeping an eye on vibrations helps prevent sudden breakdowns, saves money on repairs, and makes sure the turbine lasts a long time.

PRINCIPLE

The basic idea behind vibration measurement is to use a sensor called a transducer to turn the movement or oscillation of a machine or structure, such a turbine, into an electrical signal.

The acceleration, velocity, or displacement of the vibration is represented by this signal. A vibration module and data collecting system then process the electrical signal to examine the vibration properties. It is possible to identify particular flaws like imbalance, misalignment, or bearing damage by examining these features in the time and frequency domains. This aids in keeping an eye on the machine's health and averting mechanical malfunctions.

Data Acquisition System

A device that converts vibration sensor readings into digital data that a computer can read and process is known as a data acquisition system.

Figure 1: DAQ System

Vibration Module

The system's vibration module is a component designed specifically to process vibration signals and ensure that the data is clear and correct.

Figure 2: Vibration Module

Transducer

A sensor that senses the turbine's vibration and converts it into an electrical signal, similar to an accelerometer.

Figure 3: Transducers

LabVIEW software

A computer application that displays vibration signals in real time and facilitates simple data analysis using tools and graphs.

EXPERIMENTAL PROCEDURE

1. The vibration measurement system, including the data acquisition device and LabVIEW software, was set up properly.
2. The transducer (accelerometer) was placed securely on the turbine guide bearing to detect vibrations.
3. The transducer measured the turbine’s vibrations and converted them into electrical signals representing velocity, acceleration, and displacement.
4. These signals were processed by the vibration module and displayed in real-time using LabVIEW software.
5. The recorded time-domain vibration data was then converted into a frequency-domain plot using FFT for detailed analysis.

OBSERVATION

1. **Drive Bearings**

   Bearing type: Deep Groove Ball Bearing

   Ball diameter (d) = 35 mm

   Pitch diameter (D_p) = (236 + 384)/2 = **310 mm**

   Contact angle (β) = 0° (assumed for deep groove ball bearings)

   Number of balls (Z) = **8**

1. **BPFO – Ball Pass Frequency Outer Race**

$$BPFO = \frac{Z}{2} \times n \times (1 - \frac{d}{D_p})$$

$$BPFO = \frac{8}{2} \times 750 \times (1 - \frac{35}{310})$$

$$BPFO = 2661.20\ CPM \sim 44.35 Hz$$

2. **BPFI – Ball Pass Frequency Inner Race**

$$BPFI = \frac{Z}{2} \times n \times (1 + \frac{d}{D_p})$$

$$BPFI = \frac{8}{2} \times 750 \times (1 + 0.112)$$

$$BPFI = 3336\ CPM \sim 55.6 Hz$$

3. **BSF – Ball Spin Frequency**

$$BSF = \frac{D_p}{2d} \times n \times (1 - \left(\frac{d}{D_p}\right)^2)$$

$$BSF = \frac{310}{2 \times 35} \times 750 \times (1 - \left(\frac{35}{310}\right)^2)$$

$$BSF = 3279.76CPM \sim 54.66Hz$$

4. FTF – Fundamental Train Frequency (Cage Frequency)

$$FTF = \frac{1}{2} \times n \times (1 - \frac{d}{D_p})$$

$$FTF = \frac{1}{2} \times 750 \times (1 - 0.112)$$

$$FTF = 333\ CPM \sim 5.55Hz$$

2. Non-Drive Bearings

Bore diameter (d) 150 mm

Outer diameter (D) 225 mm

Width (B) 35 mm

Approx. roller diameter (we'll use average of E & F) (205.5 + 169.5)/2 = 187.5 mm

Number of rollers (Z) 10

Pitch diameter (D_p) Approx. center between inner & outer races: $D_p \approx (150 + 225)/2 = 187.5$ mm

Contact angle (β) 0° **(assuming radial bearing)**

1. BPFO – Ball Pass Frequency Outer Race

$$BPFO = \frac{Z}{2} \times n \times (1 - \frac{d}{D_p})$$

$$BPFO = \frac{10}{2} \times 750 \times (1 - \frac{18}{187.5})$$

$$BPFO = 3390\ CPM \sim 56.5Hz$$

2. BPFI – Ball Pass Frequency Inner Race

$$BPFI = \frac{Z}{2} \times n \times (1 + \frac{d}{D_p})$$

$$BPFI = \frac{10}{2} \times 750 \times (1 + 0.972)$$

$$BPFI = 7395\ CPM \sim 123.25Hz$$

3. BSF – Ball Spin Frequency

$$BSF = \frac{D_p}{2d} \times n \times (1 - \left(\frac{d}{D_p}\right)^2)$$

$$BSF = \frac{187.5}{2 \times 18} \times 750 \times (1 - \left(\frac{18}{187.5}\right)^2)$$

$$BSF = 3870.25\ CPM \sim 64.50Hz$$

4. FTF – Fundamental Train Frequency (Cage Frequency)

$$FTF = \frac{1}{2} \times n \times (1 - \frac{d}{D_p})$$

$$FTF = \frac{1}{2} \times 750 \times (1 - 0.972)$$

$$FTF = 10.5\ CPM \sim 0.175Hz$$

$$Turbine\ Frequency = \frac{RPM}{60}$$

$$Turbine\ Frequency = \frac{750}{60}$$

$$Turbine\ Frequency = 12.5\ Hz$$

From the graph the following data are found and calculated.

=== Displacement Analysis ===

RMS Acceleration : 1.4722 m/s²

RMS Velocity : 0.009421 m/s

Peak-to-Peak Displacement: 112938.31 µm

=== Frequency Analysis ===

Bearing Frequency Marked at: 75.0 Hz

BPFO Frequency Marked at: 44.35 Hz

BPFI Frequency Marked at: 55.6 Hz

BSF Frequency Marked at: 54.66 Hz

FTF Frequency Marked at: 5.55 Hz

Turbine Frequency Marked at: 12.5 Hz

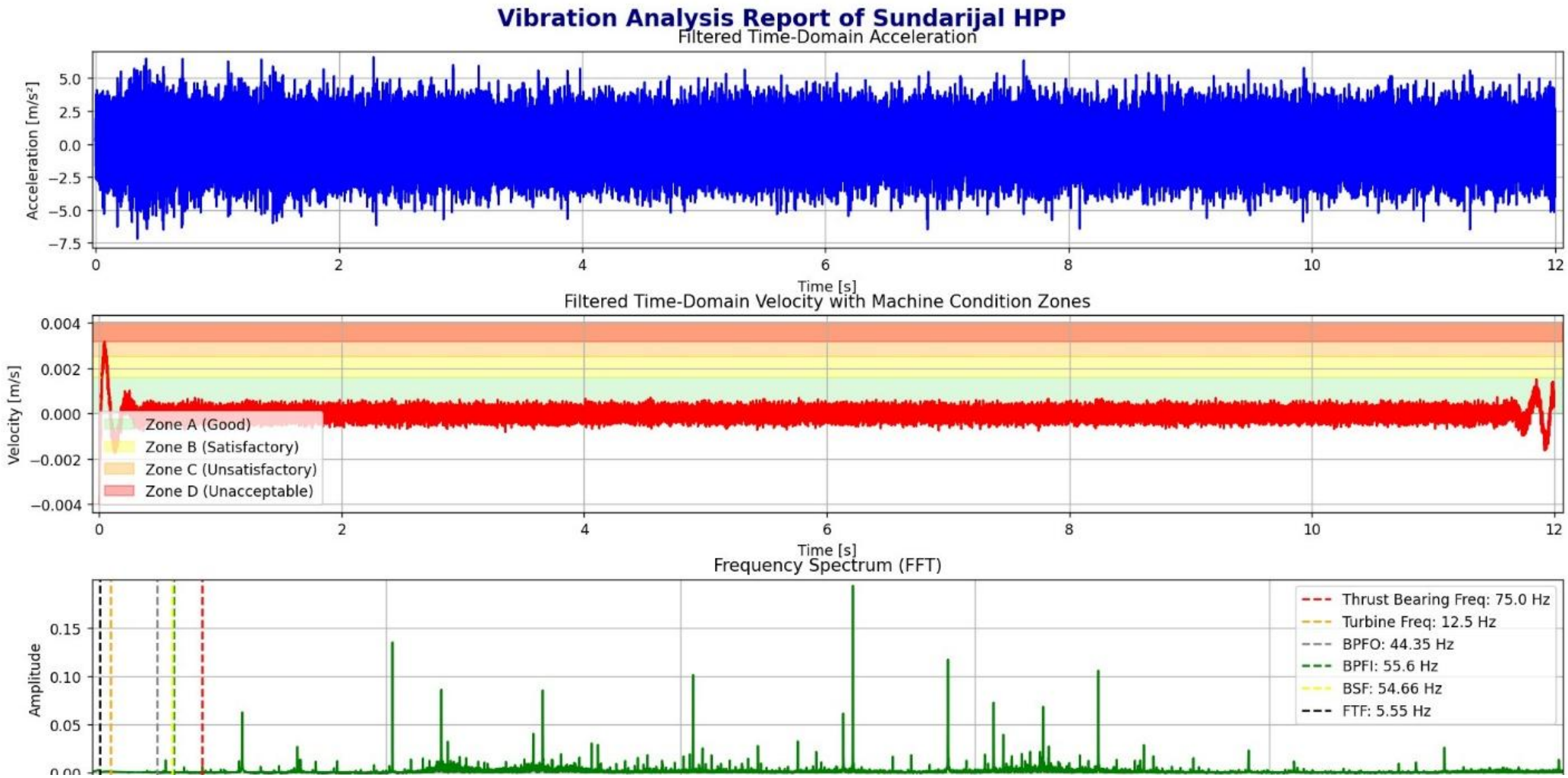


Figure 4: Vibrational Analysis of Sundarijal HPP

RESULT AND DISCUSSION

Based on the time-domain and frequency-domain analysis from the graph, it is concluded that the vibrations fall within Zone A and Zone B according to ISO 10816-5 standards. This indicates that the bearing and turbine bucket are in good condition. The maximum displacement velocity observed is 0.009 m/s, with the highest peak value recorded within acceptable limits.

CONCLUSION

The vibration analysis of Sundarijal Hydropower plant was done successfully using tools like FFT analysis, Python coding, and LabView Software, and found to be in the safe zone compared with ISO Standard 10816-5 with horizontal machine sets with bearing and moving with greater than 300rpm

ESTIMATION OF THE PLANT EFFICIENCY OF SUNDARIJAL HYDROPOWER PLANT AT DIFFERENT LOAD CONDITIONS

AIM:

1. To calculate the efficiency of a hydropower plant under different load conditions.
2. To apply the theoretical power calculation formula in the field to study discharge, head, and power.

MATERIAL REQUIRED:

1. Flow Measurement Device
2. Pressure Head Measuring Device

THEORY:

The estimation of plant efficiency at different load conditions helps us understand how effectively a hydropower plant converts the energy of flowing water into electricity under varying power demands. The plant does not always operate at full capacity; sometimes it runs at lower or medium loads depending on the electricity needs. By examining efficiency at these different loads, we can observe how performance changes. This is done by comparing the input energy from the water (based on flow and pressure) with the electrical energy generated. Such studies are important for optimizing plant operation, reducing energy losses, and ensuring reliable and efficient power generation.

DESCRIPTION:

To measure efficiency of any hydropower we use formula listed below:

$$Efficiency\ (\eta) = \frac{Output\ Power}{Input\ Power} \times 100$$

Output Power = Output power of the generator on a specific flow and head condition

Input Power= $\rho \times g \times Q \times H$

Where,

$\rho = 1000 Kg/m^2$

$g=9.81 m/s^2$

Q= Discharge

H= Pressure Head

OBSERVATION:

At Part Load Condition:

Output Power = 325000 Watt

Flowrate = 192 l/s (Measured by Ultrasonic Sensor)

Pressure = 19.91 bar

Head = 203m

At Full Load Condition:

Output Power = 495000 Watt

Flowrate = 292 l/s (Measured by Ultrasonic Sensor)

Pressure = 18.89 bar

Head = 192.67 m

CALCULATION:

At Part Load Condition:

Input Power= $\rho \times g \times Q \times H$

Input Power= $1000 \times 9.8 \times \frac{192}{1000} \times 203$ = 381964.8 Watt

$$Efficiency\ (\eta) = \frac{Output\ Power}{Input\ Power} \times 100$$

$$Efficiency\ (\eta) = \frac{325000}{381964.8} \times 100$$

= 84.99%

At Full Load Condition:

Input Power= $\rho \times g \times Q \times H$

Input Power= $1000 \times 9.8 \times \frac{292}{1000} \times 192.67$ = 551344.472 Watt

$$Efficiency\ (\eta) = \frac{Output\ Power}{Input\ Power} \times 100$$

$$Efficiency\ (\eta) = \frac{495000}{551344.472} \times 100$$

= 89.78%

RESULT AND DISCUSSION:

The efficiency of the plant was found to be 84.99% at part load and 89.78% at full load. This shows that the plant works more efficiently when it is operating at full capacity. At part load, some energy is not used as effectively, which is common in hydropower systems. The small difference in efficiency (around 4.79%) suggests that while the plant still performs well under lower loads, it reaches its best performance when running at full load. These results help highlight the importance of operating the plant as close to full capacity as possible to get the most energy output from the water used.

CONCLUSIONS:

From the study, it is clear that the efficiency of the plant improves with increased load. The plant achieved 84.99% efficiency at part load and 89.78% at full load, showing that it performs better when operating closer to its full capacity. This indicates that to maximize energy output and reduce losses, the plant should be operated at or near full load whenever possible. Understanding efficiency at different load conditions is useful for optimizing performance, improving energy management, and ensuring more reliable and cost-effective operation of the plant.

FLOW MEASUREMENT OF THE RIVER USING THE SALT DILUTION METHOD

AIM

1. To demonstrate the use of the Conductive Meter
2. To determine the discharge of the Sundarijal River.

MATERIAL REQUIRED:

1. Conductive Meter
2. Salt
3. Bucket
4. Flask with Scaling
5. Stopwatch
6. Video Recorder

THEORY:

The salt dilution method is a reliable and widely used technique for measuring river discharge, especially in small or turbulent streams where traditional methods are difficult to apply. It is based on the principle of continuity: when a known amount of salt (typically sodium chloride) is added at an upstream point, it acts as a tracer that mixes uniformly with the river flow. A conductivity meter placed at a downstream location—where the salt is fully mixed—records changes in electrical conductivity over time as the salt passes through.

By analyzing the dilution of the salt, the river discharge can be calculated using the formula:

$$Q = \frac{M}{\int C(t)dt}$$

where Q is the river discharge (m^3/s), M is the mass of salt added (kg), and C(t) is the salt concentration at time t (mg/L). The accuracy of this method depends on proper mixing of the salt, stable background conductivity, correct meter placement, and accurate measurements.

Principle:

- The technique is based on the principle that a given amount of salt is diluted more by a large amount of water than by a small amount.
- The salt dilution method involves injecting (inserting) a known amount of salt into a stream.

- The salt acts as a tracer to measure the discharge.
- The concentration of dissolved salt is measured downstream at a point

  where it has fully mixed with the stream water.

CONDUCTIVITY METER:

A conductivity meter is an instrument used to measure the electrical conductivity of a solution, which indicates the concentration of dissolved ions. It is widely used in hydrology, environmental monitoring, water quality assessment, and industrial applications.

Principle of Operation

- Conductivity meters work by passing an electrical current through a solution and measuring how easily the current flows.
- The presence of dissolved salts and ions increases conductivity, allowing the meter to determine the concentration of dissolved substances.
- The measurement is typically expressed in microsiemens per centimeter (µS/cm) or millisiemens per centimeter (mS/cm).

Figure 1: Conductivity Meter

FORMULAE:

$$Q = \frac{S}{Cal * (\sum \boldsymbol{C(t)} - (\boldsymbol{N} * \boldsymbol{C0})) * \boldsymbol{T}}$$

Where,

Q= Discharge (m^3/s)

S= Amount of salt injected in river.

Cal= Calibration factor

C(t)= Measure conducting values

C0= Base level value

N= Number of reading values

T= Time

PROCEDURE:

Calibration Procedure:

- Prepare a salt solution using 3 g of salt per 1000 ml of distilled water, using the same salt intended for injection.
- Measure exactly 500 ml of stream water and pour it into a clean 1-liter jug or beaker.
- Turn on the conductivity meter and insert its sensor into the jug.
- Record the initial conductivity on the back of the data input sheet.
- Add 1 ml of the salt solution to the jug and stir thoroughly.
- Once the conductivity reading stabilizes, record the new value.
- Repeat the steps (adding 1 ml of salt solution each time and recording) until the total volume reaches 510 ml.

Discharge Measurement Procedure with Salt

- Prepare separate plastic bags containing approximately 50 g, 100 g, and 500 g of salt, and weigh them to within ±1%. Label each bag with its exact weight using a permanent marker.
- Complete the basic information section at the top of the data input sheet.
- If possible, determine the calibration factor in advance and note it.
- Fix the conductivity meter sensor in the main flow line of the stream.
- Make a rough estimate of the stream's discharge.
- Dissolve the appropriate amount of salt (2–5 kg per m^3/s of flow) in the bucket and note the exact amount on the data input sheet.

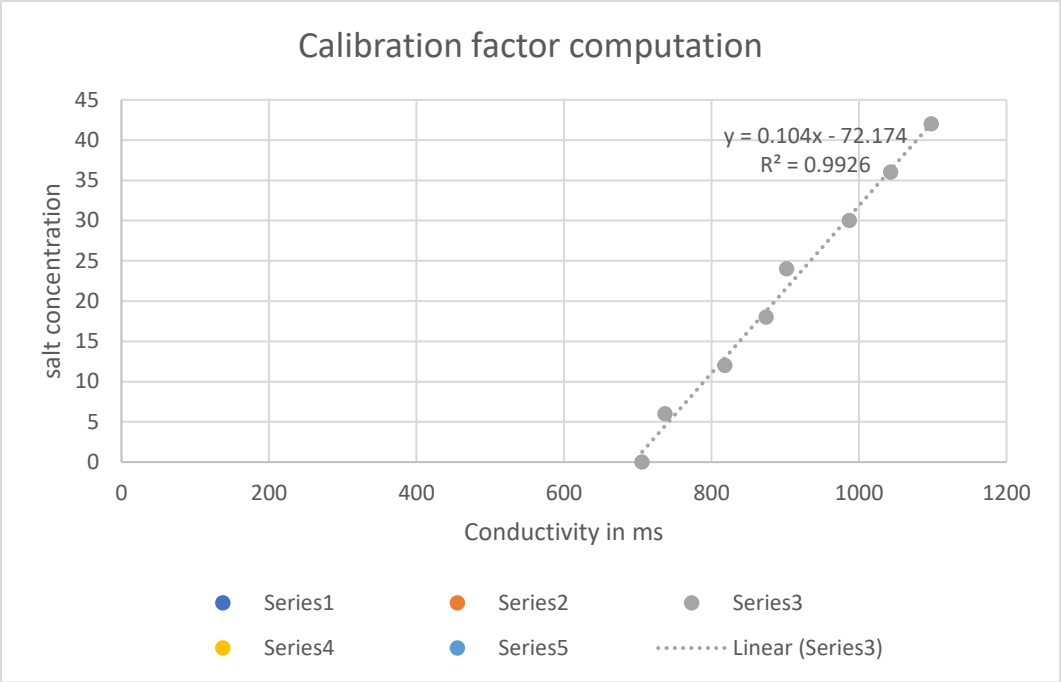
Measurement Steps

- Turn on the conductivity meter.
- Record the background conductivity on the data sheet.
- Inject the salt solution into the stream by slowly pouring it into the main flow upstream. Rinse the bucket with upstream water and pour the rinse water into the stream as well.
- When the salt reaches the measurement point, start recording the time and conductivity values at 5-second intervals.

OBSERVATION:

Calibration factor:

Calibration Sheet for Salt Dilution Method		
Sundarijal Khola		
Volume of water in jar (ml)	Measured conductivity (I s/cm)	Salt Concentration (mg/l)
500	706	0
501	737	6
502	818	12
503	874	18
504	902	24
505	987	30
506	1043	36
507	1098	42

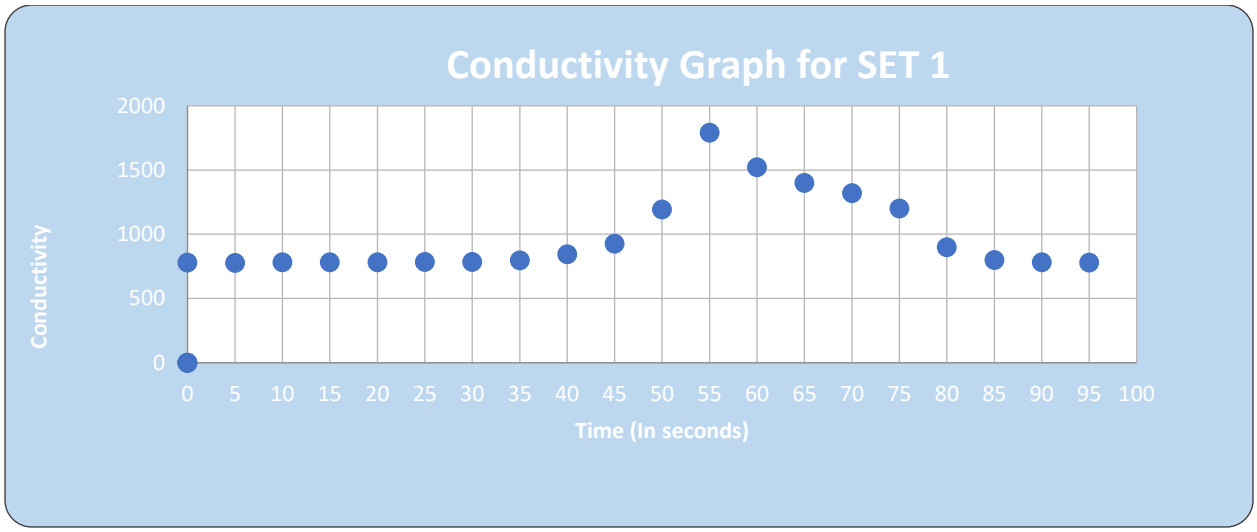


Calibration Factor found to be 0.104

Discharge Calculation:

Station no.	Sundarijal
Date	15/5/2025
Weather	
Gauge Height	
At beginning (mm)	
Time	
At beginning	
Sampeling Interval (s)	5
Distance	
Injection point	
Mixing length (m)	25
Injected salt (mg)	1000000
Other	
Water Temperature ^{0}C	22.3
Calibration coefficient (CAL)	0.104

Kali Khola			
SN	**Time**	**Conductivity (µS/cm)**	
1	0	780	
2	5	777	
3	10	782	
4	15	783	
5	20	783	
6	25	785	
7	30	785	
8	35	796	
9	40	845	
10	45	926	
11	50	1192	
12	55	1791	
13	60	1521	
14	65	1400	
15	70	1320	
16	75	1200	
17	80	900	
18	85	800	
19	90	781	
20	95	780	
		Sum (SigCt)	19727
	N		20
	C_0		791
	$N*C_0$		15820.00
	Sum - $N*C_0$		3907.00
	calibration factor		0.104 in litre/s
	Discharge, Q		492.213 in litre/s **0.492** in m^3/s



PRECAUTION AND MAINTAINENCE:

1. **Accurate Salt Measurement:** Weigh the exact amount of salt to be injected using a precise scale (±1% accuracy).
2. **Proper Salt Dissolution:** Fully dissolve the measured salt in stream water before injecting it into the flow.
3. **Correct Meter Placement:** Place the conductivity meter downstream where the salt is fully mixed with the stream water.
4. **Stable Background Conductivity:** Measure and confirm the natural conductivity of the stream before salt injection to use as a reference.
5. **Consistent Data Recording:** Start recording conductivity values just before the salt cloud arrives and continue at regular intervals until readings return to the original level.

CONCLUSION:

The salt dilution method worked well for measuring the stream's flow. We found that the discharge was 0.492 m^3/s, which shows that this method is suitable for small to medium rivers, especially in hilly or hard-to-reach areas. It's easy to use, doesn't need expensive equipment, and gives good results when done carefully. By making sure the salt was mixed properly, the meter was calibrated, and the readings were taken correctly, we got a reliable measurement of the river's flow.

FLOW MEASUREMENT OF THE RIVER USING THE AREA VELOCITY METHOD

AIM

1. To demonstrate the use of the Gurley Current Meter
2. To determine the discharge of the tailrace of Sundarijal Hydropower Plant.

Material Required:

1. Current Meter
2. Rope
3. Marker
4. Tripod containing propeller cup with sensor
5. Stopwatch

THEORY:

The Gurley Current Meter is used to measure the velocity of flowing water and calculate discharge. When the meter is placed in a stream, the water's force causes a rotating element to spin at a speed proportional to the velocity of the current. By knowing the velocity at different points and integrating across the cross-sectional area of the stream, the total discharge can be determined. The meter's readings are corrected using calibration constants that account for mechanical and hydraulic factors.

GURLEY CURRENT METER:

A Gurley Current Meter consists of:

1. A rotating cup or propeller driven by the flow of water.
2. A shaft and bearings to ensure smooth rotation.
3. A counter or electronic pulse system to record the number of rotations.
4. A suspension system to hold the meter at the desired depth in the water.

The rotating element of the Gurley Current Meter is designed to spin at a speed directly related to the velocity of the water flow. By placing the meter at various depths and positions in a channel, an accurate measurement of water velocity can be obtained. Using these velocity readings and the

cross-sectional area of the stream, the discharge can be calculated. Calibration ensures that the measured velocity corresponds accurately to real flow conditions

DESCRIPTION:

The Gurley Current Meter typically comes with several essential components to ensure accurate measurement and examination of water flow. These components include:

Figure 1: Current Meter and Rotating Cup or Propeller

1. **Meter Body**
   - The main structure houses the rotating element, bearings, and support mechanisms.
2. **Rotating Cup or Propeller**
   - Spins in response to water flow to measure velocity.
3. **Counter or Pulse Recorder**
   - Records the number of rotations over a set period to determine water speed.
4. **Calibration Chart**
   - Used to convert rotation count into actual velocity based on factory calibration.
5. **Suspension System**

- Includes rods, cables, or a wading device to hold the meter in position within the stream.

6. **Weight System**
   - Ensures the meter remains stable underwater during measurements.
7. **Carrying Case**
   - Protects the meter and its components during transport and storage.
8. **Display or Data Logger**
   - Some modern versions come with electronic displays or data loggers for real-time readings and recording.
9. **Adjustable Mounting Bracket**
   - Helps secure the meter in different environments, like open channels or pipes.

EXPERIMENTAL PROCEDURE:

1. **Select a Suitable River Section**
   - Choose a straight and stable part of the river with uniform flow.
   - Avoid areas with excessive boulders, bends, or uneven velocity distribution.
2. **Measure the River's Cross-Section**
   - Determine the width of the river using a measuring tape.
3. **Segment the River**
   - Divide the river width into 1/15 to 1/20 segments.
   - Ensure that the discharge in each segment does not exceed 10% of the total discharge.
4. **Mark the Segmented Sections**
   - Mark the segments using stakes and flags.
5. **Measure Depth for Each Segment**
   - Record depth measurements at different points within each segment and calculate the average depth.
6. **Define Cross-Sectional Areas**
   - The two edges of the river (where the current meter is placed) should be considered triangular sections.
   - All other segments should be treated as rectangular sections for discharge calculations.
7. **Submerge the Meter body and rotating part**
   - Dip the meter body, ensuring the rotating component with the sensor is positioned perpendicular to the river flow and each marking segment.
   - Make sure the cup does not rotate while submerging to avoid errors.
8. **Record Velocity Readings**
   - After submersion, allow the cup or propeller to rotate naturally with the river's flow.
   - Record the velocity of the cup within a certain time (50 seconds)
   - Note the velocity shown on the current meter for each segment.
9. **Repeat for All Segments**
   - Move the current meter systematically to each marked segment.

- Ensure proper placement and submersion at each segment, then record velocity readings.

OBSERVATION:

Location: Sundarijal Hydropower Plant							
Instrument: Gurley Current Meter						**River Width:** 0.9m	
S.N	**Width**	**Depth (d)**	**0.4d**	**Velocity m/s**	**Area m^2**	**Discharge**	
0							
1	0.3	0.49	0.196	0.92	0.147	0.13524	
2	0.6	0.49	0.196	0.84	0.294	0.24696	
3	0.3	0.49	0.196	1.01	0.147	0.14847	
4	0.6	0.49	0.196	0.75	0.294	0.2205	
5	0.3	0.49	0.196	0.84	0.147	0.12348	
6	0.6	0.49	0.196	0.92	0.294	0.27048	
7	0.3	0.49	0.196	0.84	0.147	0.12348	
8	0.6	0.49	0.196	0.84	0.294	0.24696	
9	0.3	0.49	0.196	0.92	0.147	0.13524	
10	0.6	0.49	0.196	0.84	0.294	0.24696	
				Average Discharge		**0.189777 m^3/s**	

PRECAUTIONS AND MAINTENANCE:

1. **Proper Placement:** Ensure the current meter is submerged perpendicularly to the river flow to avoid inaccurate readings.
2. **Avoid Obstructions:** Conduct measurements in areas free from turbulence, boulders, and vegetation that could interfere with flow.
3. **Stable Holding:** Secure the suspension system properly to prevent meter drift or tilting during measurement.
4. **Check Calibration:** Always verify the calibration chart or constants before taking measurements to ensure accurate results.
5. **Prevent External Errors:** Ensure the cup or propeller doesn't rotate while dipping the meter into the water to avoid false readings.
6. **Repeat Measurements:** Take multiple readings at different depths and positions in each segment to minimize error.
7. **Clean After Use:** Wash the meter with clean freshwater after every measurement to remove debris and prevent rust.

CONCLUSION:

The Gurley Current Meter provides an effective method for measuring water velocity and determining discharge in open channels. By carefully selecting a stable river section, segmenting the width, and taking multiple depth readings, accurate flow calculations can be achieved. Proper

placement and careful monitoring of the rotating component ensure precise velocity readings, which are then used to calculate discharge for each segment. In this experiment discharge of the specific place of the open channel is found to be **0.189777 m^3/s.**

Report of Hydro Lab: Pulchwok, Lalitpur

The water flow system for the Middle Kali Gandaki Hydroelectric Project, which generates 66.30 MW of energy, is the subject of this research. Being a run-of-river project, it doesn't store a lot of water; instead, it harnesses the river's natural flow. The investigation was conducted at Pulchowk, Nepal's Hydro Lab. Researchers constructed a small-scale model at a 1:40 ratio to the real project in order to comprehend how water behaves in the latter. This model assisted in simulating the flow of the river, the entry of water into the system, the movement of sediments, and the removal of undesirable elements by the basin. The system's response to floods was also tested by the model. Using specific calculations, the scaled-down model's flood flow was lowered to roughly 0.047 cubic meters per second, compared to the largest anticipated flood of 468 cubic meters per second in a year. According to a mathematical rule based on the model size, the amount of flowing water in physical modeling is substantially less than in the actual project.

Figure 1: Prototype Board of Model

The model was constructed to closely mimic the flow of water in the actual river. In order to help keep silt out, it had an intake structure that was angled. Heavy particles naturally landed on the other side of the intake due to the sloped approach; this mechanism is known as hydraulic sediment exclusion.

Large stones, however, tended to gather close to the entry gate. A unique boulder flushing gate (Gate 1) was installed to deal with this and get rid of them. The water went into a sequence of desilting basins that caught smaller particles once the boulders were flushed out. Each of the six basins had a curved curvature that increased with time.

The water was extremely turbulent at first, which made it more difficult for sediments to settle.

However, the flow steadily calmed as it passed through the basins. Water flowed smoothly enough for tiny particles to settle properly by the time it reached the sixth basin.

Figure 2: Lecture Seminar

When compared to conventional pond-based projects, this method offers significant advantages. For instance, sediment accumulation has caused the Kulekhani reservoir, which lacks a cleaning mechanism, to lose almost 70% of its capacity. The Middle Kali Gandaki project, on the other hand, minimizes sediment entry by using an inclined intake and flushing system. The amount of water flowing in the river determines how much energy is produced because this is a run-of-river facility. Power production falls with low water flow and increases with high water flow. Because of this, safeguarding the turbine against silt damage is crucial. Even microscopic silt particles can erode turbine components at high pressure, particularly in Francis turbines. The project has a methodical flushing system that runs from the intake to the desilting basins to address this problem. The amount of sediment that reaches the powerhouse is significantly decreased as a result. Proper sediment management prolongs the turbine's life and keeps it operating effectively. Regular sediment removal is also made possible by the well-designed basins and angled intake, which require less physical labor.

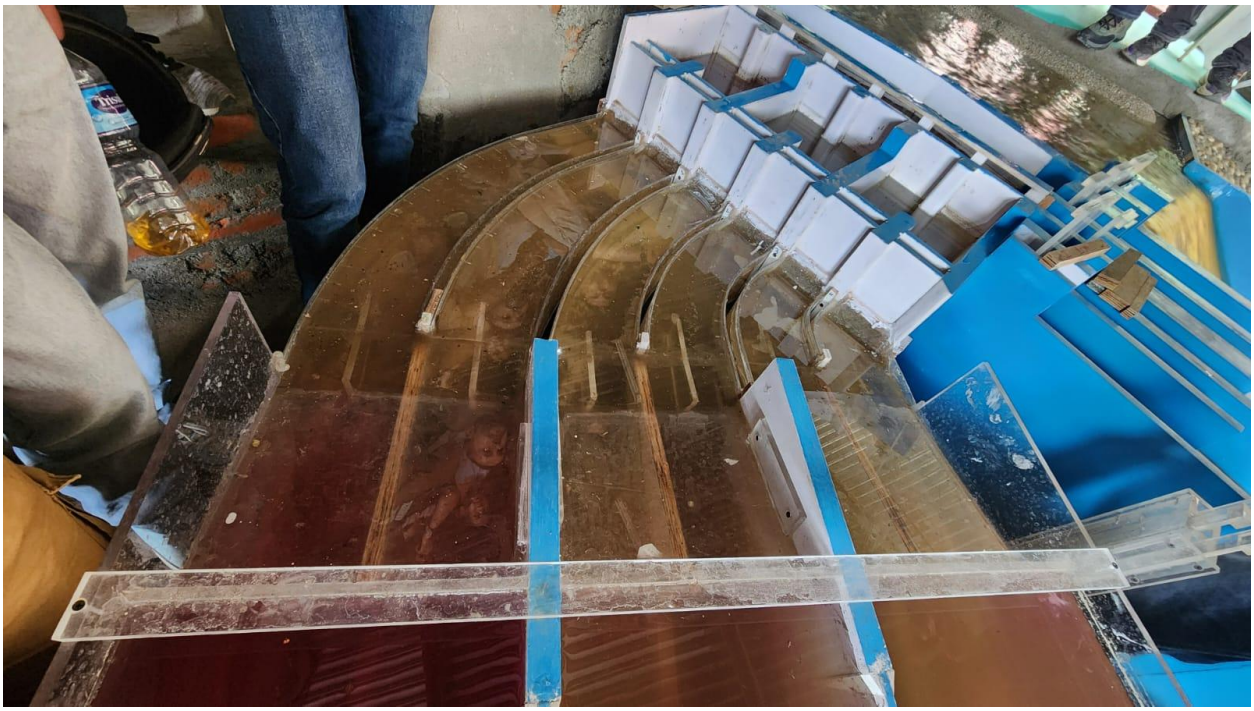


Figure 3: Deselting basin with different flow

The study concludes that the design is highly effective in preventing sedimentation, safeguarding turbine components, and maintaining consistent energy output in a run-of-river system. The scaled-down model demonstrated that sediment issues can be significantly decreased with routine flushing gate operation and adequate maintenance.

As a result, the Middle Kali Gandaki Project is both effective and long-term viable. From the intake structure to the boulder flushing gates and the precisely constructed desilting basins, every component of the hydraulic system is made to keep sediment out and guarantee the power equipment remains in good working order.

ULTRASONIC FLOW SENSOR

AIM

1. To demonstrate the use of ultrasonic flow sensor
2. To measure the penstock discharge of specific hydropower.

MATERIAL REQUIRED

1. Ultrasonic flow sensor
2. Glue
3. Ultrasonic thickness meter
4. Measuring tape

THEORY

Without coming into contact with the pipe, one can use ultrasonic flow measurement to determine how quickly a liquid passes through it. It makes use of too-high sound waves for human hearing. The IFX P200 Isoflux sensor measures the time it takes for sound waves to pass through liquid. Water and other clean liquids are the ideal candidates for this technique. The speed at which a liquid moves through a conduit can be determined using the transit-time principle. It makes use of two tiny instruments known as ultrasonic transducers, which are positioned outside the pipe. Sound waves are sent by one in the same direction as the liquid's flow and by the other in the opposite way.

The sound wave that travels with the flow gets to the other side sooner than the one that travels against it because the liquid is moving. We can determine how quickly the liquid is travelling inside the pipe by measuring the time difference between these two sound waves.

$$\Delta t = t_{up} - t_{down}$$

Using this time difference, the average flow velocity v in the pipe is calculated using:

$$v = \frac{L}{2\cos(\theta)} \times \frac{\Delta t}{t_{up} - t_{down}}$$

Where:

- L is the distance between the transducers (654 mm in this experiment)
- θ is the angle between the ultrasonic path and the flow direction
- $t_{up} - t_{down}$ are the upstream and downstream transit times

Once the average velocity is determined, the volumetric flow rate Q is found by multiplying the velocity by the internal cross-sectional area A of the pipe:

$$Q = A \times v$$

This approach consistently produces the same outcomes and is highly dependable. Using numerous sound routes, such as the 4-pass mode in this experiment, makes it even more effective. By reducing flow disruptions, these additional pathways improve measurement accuracy. Large pipes, such as the 1.49-meter-wide carbon steel pipe employed here, benefit greatly from this. Because they affix to the exterior of the pipe, ultrasonic sensors such as the IFX P200 are excellent. They are not impacted by pressure or rough pipe surfaces, nor do they come into contact with the liquid or halt its flow.

Ultrasonic flow meter

Using sound waves that are inaudible to humans, an ultrasonic flow meter measures the speed at which a liquid passes through a pipe. Both sending sound waves with and against the flow is how it operates. The sound wave that travels with the flow gets to the other side sooner than the one that travels against it because the liquid is moving. The meter determines how much liquid is passing through by detecting this time difference and calculating the liquid's flow rate.

These meters don't come into contact with the liquid inside the pipe because they are attached to the outside. They function best with clean liquids in closed pipelines and are incredibly accurate.

Figure 1: Ultrasonic Flow Meter

Ultrasonic thickness meter

Figure 2: Ultrasonic Thickness Meter

A pipe wall or other solid object's thickness can be measured with an ultrasonic thickness meter. It emits a sound wave, which the material's far side reflects back. The device determines the thickness by measuring the duration of this bounce and utilizing the known sound speed in that material. This instrument is frequently used to inspect metal for corrosion or other deterioration without breaking or cutting it.

EXPERIMENTAL PROCEDURE

1. A straight section of the pipe was selected for measurement.
2. The pipe material, outer diameter, and wall thickness were noted.
3. The fluid type and temperature were recorded.
4. The IFX P200 flow meter was powered on.
5. Pipe and fluid parameters were entered into the meter.
6. The measurement mode was set to 4-pass.
7. Coupling gel was applied to the transducers.

8. The transducers were mounted with a spacing of 654 mm.
9. Proper alignment of the transducers was ensured according to the meter's guidelines.
10. Signal strength was checked and confirmed to be adequate.
11. Flow measurement was initiated.
12. The measured flow rate was recorded.
13. The meter was powered off after the measurement.
14. The transducers were removed and cleaned.

OBSERVATION

Thickness of the pipe using an ultrasonic thickness meter = 10.1 mm

Pipe outer diameter = 1510mm

Fluid temperature = 13°C

Sensor setup = 4 pass mode

Pipe material = Carbon steel

We can also observe flow rate theoretically using the above formula, but in this experiment IFX P200 Isoflux ultrasonic flowmeter was set up using the above parameter and found a discharge of 190L/s, which is equal to 0.190m^3/s

RESULT AND DISCUSSION

An ultrasonic flow meter was used to measure the water flow in a big steel pipe. It was discovered that the flow rate was 190 liters per second. The pipe's interior width was 1490 mm, with its exterior measuring 1510 mm in breadth and its wall measuring 10 mm in thickness. This indicated that the water was moving at a speed of roughly 0.109 meters per second. By averaging several data, the 4-pass approach employed by the sensor increased measurement accuracy. The meter's settings included the water's temperature, which was 13°C. The outcomes demonstrated the effectiveness of the ultrasonic technique and were consistent with what was anticipated for a hydropower pipe with a constant flow.

CONCLUSION

The experiment effectively illustrated how to measure the water flow rate through a large-diameter carbon steel penstock in a hydropower system using the IFX P200 Isoflux ultrasonic flow meter.

Without interfering with the flow, the non-invasive, transit-time-based measurement technique produced precise and trustworthy data. For the specified pipe dimensions and operating circumstances, the computed flow velocity and the actual flow rate of 190 L/s were in good agreement. All things considered, the ultrasonic flow meter worked well for measuring flow in industrial settings with big pipelines and pure fluids.

NEPAL YANTRA SHALA ENERGY (NYSE) EXCURSION

1. Introduction

One of Nepal's leading companies in the production of hydropower equipment is Nepal Yantra Shala Energy (NYSE). Since its founding in 1971, NYSE has been instrumental in increasing Nepal's ability to develop dependable, location-specific micro-hydropower technology. It has grown to be a vital component in the nation's endeavor to increase access to renewable energy, especially in rural and off-grid areas, over the years.

The company focuses on designing, developing, and building small-scale hydro turbines, specifically Pelton, Crossflow, and Propeller types. Each turbine is tailored to the site's particular hydraulic parameters, including head, flow, and sediment load. Because of its capacity to produce such vital parts domestically, NYSE has become a major force in encouraging the growth of hydropower in Nepal, lowering reliance on imports, and advancing sustainable energy practices.

Casting, machining, balancing, and performance testing of turbine components are all done in-house at NYSE. The business guarantees quality control at every level of manufacturing with this integrated approach. One of its main initiatives is the local development of Turgo turbines, which is being carried out in cooperation with Kathmandu University and energy-focused organizations like PEEDA and HPNET. It also places a strong emphasis on product innovation.

In addition to manufacturing, NYSE provides technical training, system installation, and engineering consulting services to the hydropower industry. Its efforts assist the larger goals of energy equity and self-reliance in addition to fortifying Nepal's hydroelectric infrastructure. NYSE continues to be a premier example of how locally based engineering solutions can meet national development goals, having operated for more than 50 years.

2. Objectives

- To observe the step-by-step fabrication of micro-hydro turbines, especially Pelton types.
- To engage with engineers and technicians to learn about challenges and best practices in small-scale turbine production.

3. Manufacturing Process Observed

a. Casting

In order to create the rough shape of turbine components, raw metal is melted and poured into specially made molds at the start of the manufacturing process. In our instance, we saw that the Pelton runner cups, essential components of impulse turbines, were cast. The runner bucket's basic geometry is formed by these first castings, which are not yet prepared for assembly. After cooling and cleaning, the castings are ready for the following refinement step. This stage is crucial because it establishes the general quality of the material and the foundation for accurate machining in the future.

b. Machining and Finishing

The rough parts are precisely machined after casting. Using lathe machines and CNC equipment, this step involves surface finishing, drilling, turning, and milling. Reaching the precise design dimensions and tolerances for the cast parts is the aim. To avoid vibrations while in operation, special attention is paid to the runner's balance, which ensures homogeneous mass distribution. Additionally, the surfaces are polished to reduce hydraulic losses and friction. This stage guarantees that every component fits together perfectly and operates dependably during assembly.

Figure: Surface Finishing

c. Assembly

The machined parts are put together into working subsystems at this point. Other turbine components, such as nozzles, guide vanes, casings, and bearings, are fitted once the Pelton runner is fixed to the shaft. Water leaks are stopped by mechanical seals, and the shaft is carefully aligned to guarantee steady rotation. This stage is essential to guaranteeing the finished turbine unit's hydraulic effectiveness and mechanical integrity.

d. Testing

The assembled turbine is checked for flow efficiency, leakage, vibration, and mechanical alignment when transferred or installed. Dry rotation inspections, hydraulic testing on a test bench, and pressure testing to find leaks are a few examples of these tests. To make sure the turbine works well with the pipeline and load configuration, site-level testing is occasionally done after the turbine is transported to the installation location, particularly in community or off-grid systems.

e. Painting and Protective Coating

Once finalized, the turbine components are coated with protective paint or anti-corrosive layers, especially if they are to be installed in high-silt or outdoor environments. This helps increase the lifespan of the machinery and protects it from rust and erosion.

f. Packaging and Dispatch

After testing, the turbines are carefully packed for transportation. Proper labeling, documentation, and handling instructions are added to ensure safe delivery to remote or mountainous project sites.

4. Manufacturing Equipment & Tools

a. Metal Foundry Setup for Casting Runner Cups

The metal foundry is where the initial shaping of turbine components begins. In this facility, raw metal is melted at high temperatures and poured into molds to create rough castings of components such as Pelton runner cups. The foundry includes:

- Crucibles for melting metal.
- Molds (often made from sand or metal) for shaping the molten metal.

- Pouring systems and cooling platforms.
  This stage is foundational, as the structural integrity and geometry of the cast parts depend on the quality of the casting process.

b. CNC Milling and Turning Machines for Precise Shaping

Computer Numerical Control (CNC) machines are used to shape the cast parts to precise dimensions. These machines allow for high-accuracy milling, drilling, and turning operations.

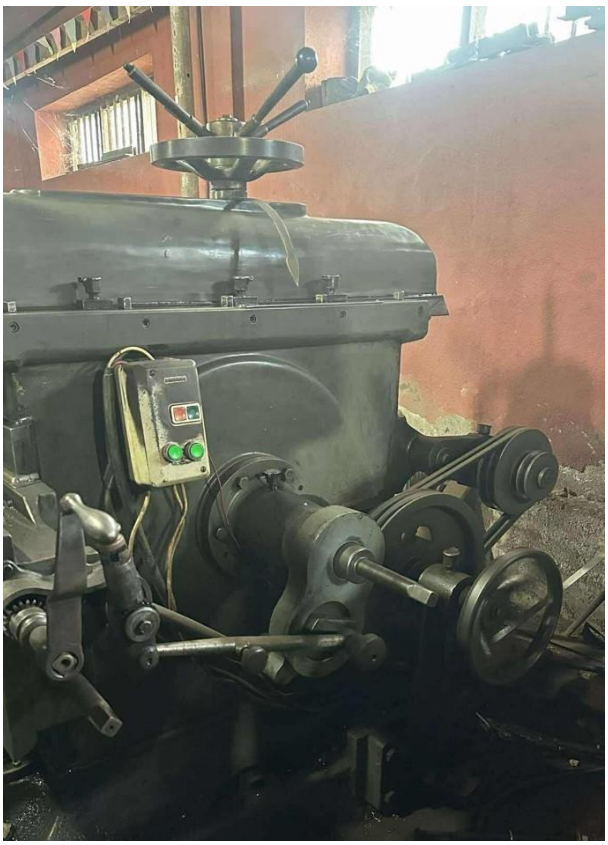


Figure: Lathe

At NYSE, CNC and manual lathes were used to:

- Create smooth, accurate surfaces on runners and shafts.
- Drill holes for mounting and alignment.
- Finish surfaces to reduce hydraulic losses.
  CNC machines significantly improve repeatability and reduce human error, making them essential for turbine production.

c. Balancing Equipment for Runner Alignment

Balancing equipment is used to ensure that rotating components such as Pelton runners are perfectly symmetrical in mass distribution. Any imbalance can cause:

- Vibration during operation.
- Premature wear of bearings and seals.
- Reduced efficiency and mechanical failure.

  Dynamic balancing machines spin the runner and detect any mass irregularities, after which material is added or removed to achieve proper balance.

d. Hand Tools for Alignment, Sealing, and Assembly

A variety of manual tools are used throughout the manufacturing process for tasks that require finesse and fine adjustment. These include:

- Torque wrenches and spanners for bolt tightening.
- Alignment gauges for shaft and bearing fitting.
- Sealant applicators and rubber hammers for leak-proof assembly.

  Though simple, these tools are essential for achieving precision and proper integration of turbine parts.

5. Interaction with NYSE Engineers

We were able to have reflective conversations with the technical team about their manufacturing processes and engineering strategy.

We entered into the problems of the casting process, more specifically for Pelton runner cups. Consistency in metal composition, shrinkage cavity prevention, and maintaining dimensional accuracy during cooling are the supreme concerns. The team outlined how defects here can lead to machining defects or imbalance in operation, and hence need to be carefully examined and put through rigorous quality control processes.

Another issue of contention was machining accuracy and balancing techniques. Engineers talked about how small discrepancies in runner bucket alignment or surface finish would drastically reduce the hydraulic efficiency of the turbine. Therefore, NYSE employs high precision machinery and carries out multiple phases of inspection while machining to ensure consistency and reliability.

Figure: Interaction with NYSE Engineers

6. Reflections & Learning Outcomes

We came away with an enhanced respect for the principle of site-specific turbine tailoring. Each project must have a unique configuration based on the head and flow of the site. This real-world experience worked to reinforce the engineering principle that hydropower performance optimization is intimately connected with context-sensitive design, rather than one-size-fits-all approaches.

The visit highlighted the value placed on accuracy and quality control in turbine manufacturing. Minute discrepancies in dimension or alignment can mean inefficiency or mechanical failure. Witnessing the attention afforded to machining, balancing, and assembly demonstrated how small mistakes become amplified into serious operational issues without correction.

Lastly, we witnessed innovation being done in a local context. NYSE, coming up with turbines that are tailor-made for Nepal's medium-head sites, illustrated to us how research-based solutions could be translated into products that could be manufactured. It illustrated to us that innovation is not the domain of large international companies but can be led by local manufacturers who are aware of local needs and constraints.

7. Conclusion

The NYSE visit provided valuable insight into the practical aspects of Nepal's micro-hydropower turbine production. The experience demonstrated the direct influence of fundamental hydraulic factors on design and production choices, including head, flow rate, and sediment load. By creating specialized turbines that satisfy various site conditions, it also emphasized the important role that regional organizations play in expanding Nepal's potential for renewable energy. All things considered, the trip developed an appreciation for engineering that is grounded in both technical precision and contextual awareness.